ENGR-095
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10/8/2026

During the ENGR-095 Microcontrollers unit, I found that I learned many applicable skills- as well as many things about where my interest in technology and engineering lie in general.

When the unit began, I had always considered coding to not be an area of my interest- and I always found I had more interest in physics in general, and had quite a well placed understanding of electricity when I learned that unit in my high school education. During this unit, we were introduced to microcontrollers and circuits designed to help assist in engineering and coding.

I had always been interested in arduinos- yet I never really had access to much electrical engineering or coding applications in my high school. We did engineering basics, however we never got a deep dive into topics, or had access to the kind of hardware that is supplied at Lehigh. We did not have a large engineering budget, and we used leftover materials from the years prior- although we did have a laser cutter, which was incredibly cool. However we did not use it very much. Despite this, I found myself incredibly interested and drawn to the process of electrical engineering.

While I am aware this segment was focusing on educating microcontrollers more so on the materials science side, explaining how they are made and how they function, I found that I was quite enamored by the breadboarding and wiring- as well as the interfacing with the pins, and the software. In general, I found myself much more interested in the assembly and hands on building of the circuits and applying the microcontrollers. When I got to Lehigh, I was planning on being a Bioengineering / Mechanical major. However, after taking some classes at Lehigh, I realized that my interests were not actually placed where I thought they were. I believe that my experience in this FYRE module, as well as my extra curricular activities played a heavy role in shaping my newfound interests and desires in college.

From this microcontroller unit, I not only learned how these pieces were assembled- but also how they interacted with practical / physical machinery and circuitry. When creating real prototypes, it's different than what you interface, or even write down to plan everything. Things change a little in the transition from theoretical to hands on (although in some cases that cannot be true, such as rocket design, for example- that has to be pretty much on the nose, as well as other critical / potentially dangerous engineering types.)

Ultimately, as far as the microcontrollers unit goes, I have learned so much from this experience in the module. I have found that this section in particular really sparked my interest in electrical engineering. I had always considered engineering to be computer engineering, software engineering, and mechanical. For some reason, I always thought electrical engineering was primarily computer engineering- however, upon realizing, I was just dead wrong- and I've found a love for circuitry through this unit. I've learned practical and conceptual skills in this unit, and have found them to be incredibly useful.